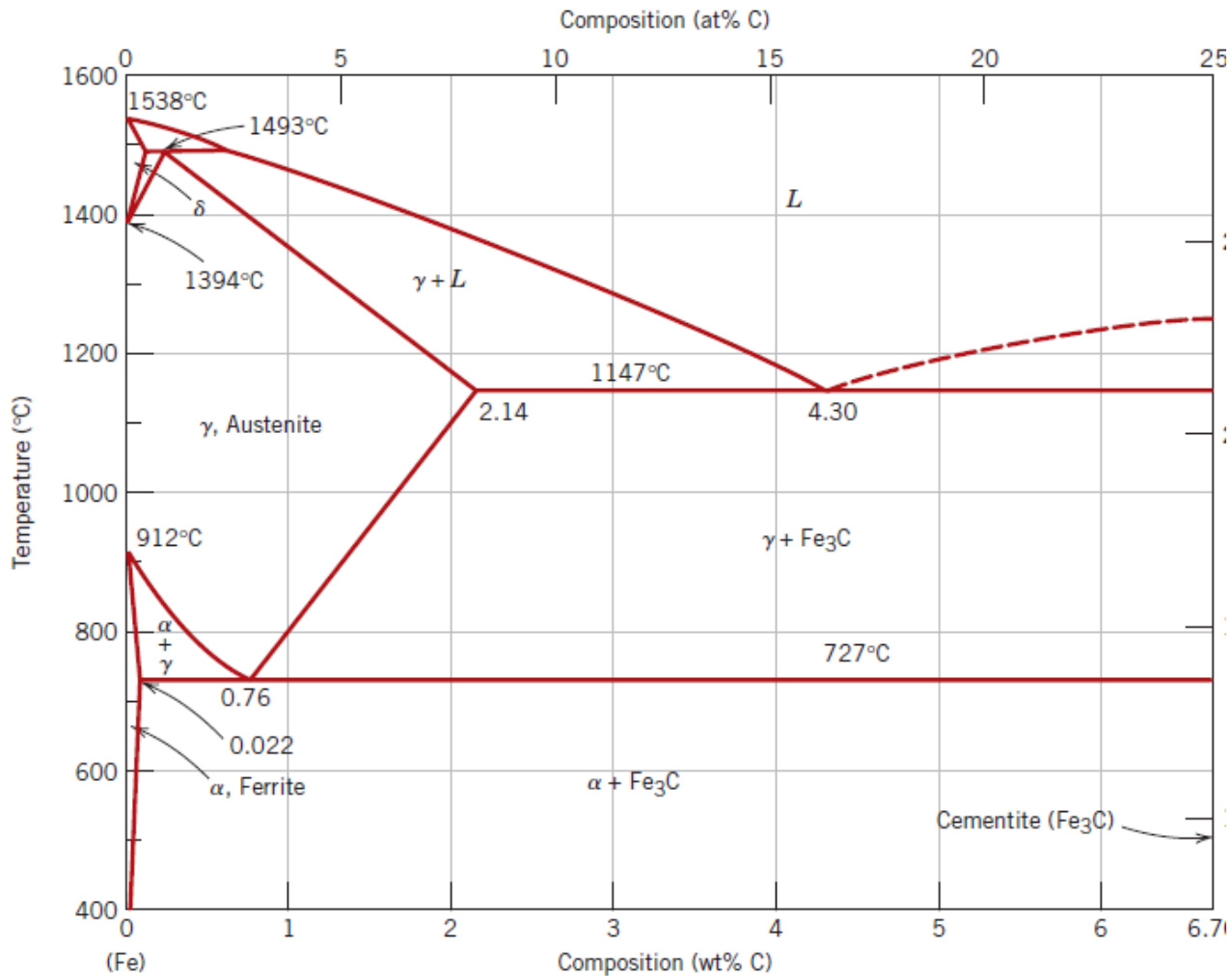


Problems on Fe-C phase diagrams



1.- For a Fe-C alloy with $C_0 = 0.5\%C$, at the following temperatures: $T_1 = (1147+\Delta T)^\circ C$; $T_2 = (1147-\Delta T)^\circ C$; $T_3 = 900^\circ C$; $T_4 = (727+\Delta T)^\circ C$; $T_5 = (727-\Delta T)^\circ C$

a) Indicate the phases present, their chemical composition and their mass fraction.

b) Indicate the micro constituents, their chemical composition and their mass fraction at T_5 .

Solutions: a) T_1, T_2 , and T_3 : $C_\gamma = 0.5\% C$, $w_\gamma = 1$; T_4 : $C_a = 0.022\% C$, $C_\gamma = 0.76\% C$, $w_a = 0.35$; T_5 : $C_a = 0.022\% C$, $C_{Fe_3C} = 6.7\% C$, $w_a = 0.93$; b) $w_{a'} = 0.35$, $w_{a,E} = 0.58$, $w_{Fe_3C} = 0.07$)

2.- For a steel of $C_0 = 0.76\%C$, at the following temperatures: $T_1 = (1147+\Delta T)^\circ C$; $T_2 = (1147-\Delta T)^\circ C$; $T_3 = 900^\circ C$; $T_4 = (727+\Delta T)^\circ C$; $T_5 = (727-\Delta T)^\circ C$

a) Indicate the phases present, their chemical composition and their mass fraction.

b) Indicate the micro constituents, their chemical composition and their mass fraction at T_5 .

Solution: a) T_1, T_2, T_3 and T_4 : $C_\gamma = 0.76\% C$, $w_\gamma = 1$; T_5 : $C_a = 0.022\% C$, $C_{Fe_3C} = 6.7\% C$, $w_a = 0.89$, $w_{Fe_3C} = 0.11$; b) pearlite $w_{a,E} = 0.89$, $w_{Fe_3C,E} = 0.11$

3.- For an alloy Fe-1.5%C, at the following temperatures: $T_1 = (1147+\Delta T)^\circ\text{C}$; $T_2 = (1147-\Delta T)^\circ\text{C}$; $T_3 = 900^\circ\text{C}$; $T_4 = (727+\Delta T)^\circ\text{C}$; $T_5 = (727-\Delta T)^\circ\text{C}$

- Indicate the phases present, their chemical composition and their mass fraction.
- Indicate the micro constituents, their chemical composition and their mass fraction at T_5 .

Solution: a) T_1 and T_2 : $C_\gamma = 1.5\% \text{ C}$, $w_\gamma = 1$; T_3 : $C_{\text{Fe}_3\text{C}} = 6.7\% \text{ C}$, $C_\gamma = 1.3\% \text{ C}$, $w_\gamma = 0.96$, $w_{\text{Fe}_3\text{C}} = 0.04$; T_4 : $C_\gamma = 0.76\% \text{ C}$, $C_{\text{Fe}_3\text{C}} = 6.7\% \text{ C}$, $w_\gamma = 0.87$, $w_{\text{Fe}_3\text{C}} = 0.13$; T_5 : $C_\alpha = 0.022\% \text{ C}$, $C_{\text{Fe}_3\text{C}} = 6.7\% \text{ C}$, $w_\alpha = 0.78$, $w_{\text{Fe}_3\text{C}} = 0.22$; b) Pearlite $w_{\text{Fe}_3\text{C}'} = 0.13$, $w_{\text{pearlite}} = 0.87$ $w_{\alpha\text{E}} = 0.78$, $w_{\text{Fe}_3\text{C,E}} = 0.09$

4.- For an alloy Fe- 4.3% of C, at the following temperatures:

$T_1 = 1147 + \Delta T^\circ\text{C}$; $T_2 = 1147 - \Delta T^\circ\text{C}$; $T_3 = 900^\circ\text{C}$; $T_4 = 727 + \Delta T^\circ\text{C}$; $T_5 = 727 - \Delta T^\circ\text{C}$

- Indicate the phases present, their chemical composition and their mass fractions.
- Indicate the micro constituents, their chemical composition and their mass fractions.

(Solutions: a) T_1 : $C_L = 4.3\% \text{ C}$, $w_L = 1$; T_2 : $C_{\text{Fe}_3\text{C}} = 6.7\% \text{ C}$, $C_\gamma = 2.14\% \text{ C}$, $w_{\gamma\text{E}} = 0.53$; T_3 : $C_{\gamma\text{E}} = 1.3\% \text{ C}$, $C_{\text{Fe}_3\text{C}} = 6.7\% \text{ C}$, $w_\gamma = 0.44$; T_4 : $C_\gamma = 0.76\% \text{ C}$, $C_{\text{Fe}_3\text{C}} = 6.7\% \text{ C}$, $w_\gamma = 0.40$; T_5 : $C_\alpha = 0.022\% \text{ C}$, $C_{\text{Fe}_3\text{C}} = 6.7\% \text{ C}$, $w_\alpha = 0.36$; b) At T_2 : $w_{\text{Fe}_3\text{C,E}} = 0.47$, $w_{\gamma\text{E}} = 0.53$)

5.- Calculate the composition of a Fe-C alloy whose composition is within the range in %C of $(2.11 < C_0 < 4.3)$. Consider temperature is $(727-\Delta T)^\circ\text{C}$ and the mass fraction $\chi_{\text{Fe}_3\text{C}}$ is 0.38.

Solution: $C_0 = 2.56\% \text{ C}$

6.- Consider a steel with a content of $C < 4.3\%$ at $(727-\Delta T)^\circ\text{C}$. In these conditions, $\chi_\alpha = 0.47$. Indicate the mass fraction of the phases at $(1147+\Delta T)^\circ\text{C}$ and $(1147-\Delta T)^\circ\text{C}$.

Solution: T_1 : $w_\gamma = 0.32$, $\chi_L = 0.68$; T_2 : $w_\gamma = 0.68$, $\chi_{\text{Fe}_3\text{C}} = 0.32$, $w_E = 0.68$, $\chi_{\gamma\text{E}} = 0.36$, $w_{\text{Fe}_3\text{C}} = 0.32$

7.- Consider an alloy of Fe-C, with content of $C > 4.3\% \text{ C}$ at $(727-\Delta T)^\circ\text{C}$. In these conditions, $\chi_{\text{Fe}_3\text{C}} = 0.73$. Indicate the mass fraction of the phases at $(1147+\Delta T)^\circ\text{C}$ and $(1147-\Delta T)^\circ\text{C}$

Solution: T_1 : $w_L = 0.75$, $w_{\text{Fe}_3\text{C}} = 0.25$; T_2 : $w_\gamma = 0.39$, $w_{\text{Fe}_3\text{C}} = 0.61$, $w_{(\text{Fe}_3\text{C})\text{p}} = 0.25$, $w_E = 0.75$, $w_{\gamma\text{E}} = 0.39$, $w_{(\text{Fe}_3\text{C})\text{E}} = 0.36$

8.- Examine closely the phase diagram Fe-C and answer the following questions.

- If there is a 5kg annealed bar of an alloy with 0.55 % C, what is the quantity (mass expressed in Kg) of pre-eutectoid ferrite, eutectoid ferrite and cementite that is contained at 726°C ?
- List the phases and microconstituents in this steel, their composition and their percentage at 600°C
- How are the carbon content and temperature at the eutectoid point modified in the diagram if Ni is added? How does this addition modify the steel's microstructure?

Solutions: a) $\alpha' = 1.43\text{kg}$; pearlite = 3,6kg; $\alpha_{\text{total}} = 4.6 \text{ kg}$; $\alpha_{\text{eutectoid}} = 3.17\text{kg}$; $\text{Fe}_3\text{C} = 0.4\text{kg}$; b) $\chi_{\alpha\text{pre}} = 0.26$; $\chi_{\text{eutectoid}} = 0.72$; $w_{\alpha\text{total}} = 0.92$; $w_{\text{Fe}_3\text{C}} = 0.08$; c) possible decrease in amount of in primary ferrite

9. Examine closely the phase diagram Fe-C answer the following questions.

- Calculate the percentage of C in an alloy that has 64.7 % of austenite and 35.5 % of cementite at 800 °C. What kind of alloy is it?
- What microconstituents does it have at 1146°C? Calculate the quantity of each one.
- How do microconstituents change when cooling down from 1146 °C to 800 °C?

Solutions: a) $C_0=3\%$; hypoeutectic cast iron; b) $w_{\gamma\text{primary}}=60.2\%$; $w_{\text{ledeburite}}=39.8\%$ $w_{\text{Fe}_3\text{C}} = 18.8\%$ $w_{\gamma\text{ledeburite}}=21\%$ $w_{\gamma\text{total}}=81.2\%$ c) Fe_3C precipitates (secondary Fe_3C); $w_{\text{Fe}_3\text{C secondary}}=16.3\%$ d

10.- Using the Fe-cementite phase diagram, indicate:

- What real alloy or alloys comply the following conditions $W_\alpha = 0.267$ at $727 - \Delta T$ °C?
- What is the maximum and minimum content (mass fraction W) of pearlite in a steel? Explain why.
- Indicate the invariant points that appear in the diagram and explain which microconstituents are formed during cooling.
- For the alloys with 1.5% C and 3% C, indicate the microconstituents and if the mass fraction of them changes on cooling in the range between 1148 °C and $727 + \Delta T$ °C.

(Solutions: a) $C_0 = 4.9\% \text{ C}$; b) Max: 100% for $C_0=0.76\% \text{ C}$; min: 0% for $C_0 < 0.022\% \text{ C}$ d) 1.5% C; T_1 : $w_\gamma = 1$, T_2 : $w_{\text{Fe}_3\text{C}} = 0.12$, $w_\gamma = 0.88$; 3% C; T_1 : $w_\gamma = 0.6$, $w_\text{L} = 0.4$, T_2 : $w_{\gamma\text{total}} = 0.62$, $w_{\text{Fe}_3\text{C total}} = 0.38$, $w_{\text{Fe}_3\text{C secondary}} = 0.20$

11.- Determine the composition in wt% of the ternary alloy H marked in the diagram below:

(Solution: 40% C + 33% A + 30%B)

